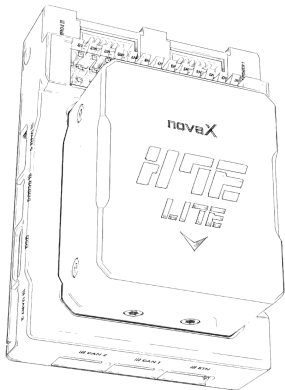


# AF-H7E Lite

## User Guide



# AF-H7E Lite Overview



Compact carrier flight controller that reuses the AF-H7E sensor module (in development). The pin definition may change before release.

# AF-H7E Lite Specifications

## • Basic Specifications

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| MCU                  | STM32H753, ARM Cortex-M7,<br>480 MHz                             |
| RAM / Flash          | 1 MB / 2 MB                                                      |
| IMU                  | ICM-42688-P, BMI088                                              |
| Secondary IMU        | ICM-20649 (optional)                                             |
| Magnetometer         | RM3100 (optional; use an<br>external compass when not<br>fitted) |
| Barometer            | 2× ICP-20100                                                     |
| Operating<br>Voltage | 4.75 – 5.7 V (Rated 5 V)                                         |
| USB Input            | 4.75 – 5.25 V                                                    |
| Power Input          | POWER 1 + POWER 2<br>(redundant, I2C power<br>modules)           |

# AF-H7E Lite Specifications

## • Basic Specifications

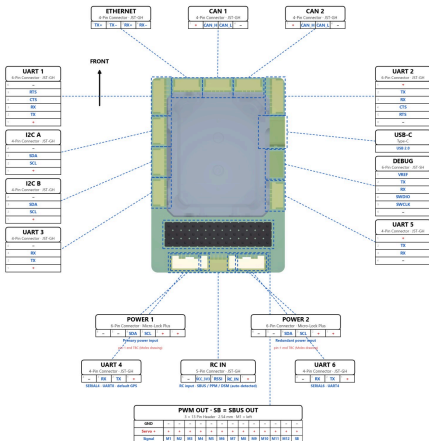
|                  |                                                          |
|------------------|----------------------------------------------------------|
| PWM Output       | 12 FMU channels (DSHOT except M7–M8) + SBUS out          |
| RC Input         | S.Bus, PPM, DSM / Spektrum (with DSM bind)               |
| RSSI Input       | Analog                                                   |
| UART             | 6 serial ports (2 with flow control) + debug console     |
| I <sup>2</sup> C | 2 ports (I2C A, I2C B) + power-module bus on POWER 1 · 2 |
| CAN              | 2 Port                                                   |
| Ethernet         | 100 Mbps x 1 Port                                        |
| Size             | 43.0 × 63.4 × 29.2 mm (concept case)                     |
| Supported F/W    | novaX ArduPilot (Copter and Plane releases)              |

# AF-H7E Lite Pinout

## • Pinout

### AF-H7E Lite — Pinout (Top View)

Concept layout - pin order as seen from above - side connectors sit on the bottom side and plug in from the edges (shown through the PCB)



# AF-H7E Lite Pinout

## • Pinout Notes

Preliminary pin definition — AF-H7E Lite is in development and connectors may change before release. On every JST connector pin 1 is the supply pin and the last pin is GND, following the Pixhawk connector standard. UART n maps to ArduPilot SERIALn: UART 1–2 default to MAVLink telemetry and UART 3–4 default to GPS. GPS modules connect to any UART or over DroneCAN; there is no dedicated GPS/safety port and no safety switch. RC IN is a 5-pin connector with the same pinout as AF-H7E and takes an SBUS, PPM or DSM receiver directly on pin 2 (protocol auto-detected, wired to the flight-controller MCU, no IO board needed); pin 1 supplies 5 V and pin 4 a switched 3.3 V for DSM / Spektrum satellite receivers, which the flight controller power-cycles to bind them. On-board without a connector: microSD card slot (logging), buzzer and RGB status LED.

# AF-H7E Lite Pinout

The + rail of the PWM header is not powered by the flight controller. The 13th header column, SB, is an SBUS output, not a PWM channel: it carries servo channels 1–16 on one wire from USART6 (SERIAL8) with the signal inversion done inside the STM32H7, so SBUS servos, SBUS-to-PWM decoders and gimbals plug in with a standard servo lead and take power from the servo rail. Outputs share rate and protocol within the timer groups M1–M4, M5 · M6 · M9 · M10, M7–M8 and M11–M12; DShot works on every group except M7–M8, whose timer has no DMA (PWM and OneShot only).

# AF-H7E Lite Pinout

POWER 1 and POWER 2 sit on the top side at the rear edge with their latch facing the rear wall, so the plug is released from outside the case through the wall opening; RC IN, UART 4 and UART 6 sit under them on the bottom side and plug in from the rear like the side connectors. The PWM header sits 1.5 mm behind the sensor module: as on AF-H7E the signal row (S) is the rearmost row and the case has a keyed comb behind it, so servo plugs only fit with S at the rear. The pinout image is a top view of the concept carrier: side connectors are mounted on the bottom side and plug in from the edges, and each table lists pins in the order they sit when seen from above. POWER 1 and POWER 2 use the same Molex Micro-Lock Plus connector as AF-H7E; their pin-1 end is still to be confirmed. POWER 1 and POWER 2 take I2C (INA2xx) power modules; a DroneCAN power module (for example 14S / 200 A) reports over CAN 1 or CAN 2 (BATT\_MONITOR 8) and needs a split cable that feeds its 5 V into POWER 1, because the CAN connectors supply 5 V rather than accept it.

# AF-H7E Lite Pin Definition

## • POWER 1

Molex Micro-Lock Plus 6P (1.25 mm) · same as AF-H7E ·  
Primary power input

| Pin | Signal | Function                                           |
|-----|--------|----------------------------------------------------|
| 1   | VCC_IN | 5 V supply input from the power module             |
| 2   | VCC_IN | 5 V supply input from the power module             |
| 3   | SCL    | Power-module I2C clock (voltage / current monitor) |
| 4   | SDA    | Power-module I2C data (voltage / current monitor)  |
| 5   | GND    | Ground                                             |
| 6   | GND    | Ground                                             |

# AF-H7E Lite Pin Definition

## • POWER 2

Molex Micro-Lock Plus 6P (1.25 mm) · same as AF-H7E ·  
Redundant power input

| Pin | Signal | Function                                           |
|-----|--------|----------------------------------------------------|
| 1   | VCC_IN | 5 V supply input from the second power module      |
| 2   | VCC_IN | 5 V supply input from the second power module      |
| 3   | SCL    | Power-module I2C clock (voltage / current monitor) |
| 4   | SDA    | Power-module I2C data (voltage / current monitor)  |
| 5   | GND    | Ground                                             |
| 6   | GND    | Ground                                             |

# AF-H7E Lite Pin Definition

## • UART 1

JST-GH 6P · SERIAL1 · UART7 · default MAVLink2

| Pin | Signal | Function                                       |
|-----|--------|------------------------------------------------|
| 1   | VCC    | 5 V output                                     |
| 2   | TX     | UART transmit (FC → device)                    |
| 3   | RX     | UART receive (device → FC)                     |
| 4   | CTS    | Clear to send — hardware flow control input    |
| 5   | RTS    | Request to send — hardware flow control output |
| 6   | GND    | Ground                                         |

# AF-H7E Lite Pin Definition

## • UART 2

JST-GH 6P · SERIAL2 · UART5 · default MAVLink2

| Pin | Signal | Function                                       |
|-----|--------|------------------------------------------------|
| 1   | VCC    | 5 V output                                     |
| 2   | TX     | UART transmit (FC → device)                    |
| 3   | RX     | UART receive (device → FC)                     |
| 4   | CTS    | Clear to send — hardware flow control input    |
| 5   | RTS    | Request to send — hardware flow control output |
| 6   | GND    | Ground                                         |

## • UART 3

JST-GH 4P · SERIAL3 · USART1 · default GPS

| Pin | Signal | Function                    |
|-----|--------|-----------------------------|
| 1   | VCC    | 5 V output                  |
| 2   | TX     | UART transmit (FC → device) |
| 3   | RX     | UART receive (device → FC)  |
| 4   | GND    | Ground                      |

# AF-H7E Lite Pin Definition

## • UART 4

JST-GH 4P · SERIAL4 · UART8 · default GPS

| Pin | Signal | Function                    |
|-----|--------|-----------------------------|
| 1   | VCC    | 5 V output                  |
| 2   | TX     | UART transmit (FC → device) |
| 3   | RX     | UART receive (device → FC)  |
| 4   | GND    | Ground                      |

## • UART 5

JST-GH 4P · SERIAL5 · USART2

| Pin | Signal | Function                    |
|-----|--------|-----------------------------|
| 1   | VCC    | 5 V output                  |
| 2   | TX     | UART transmit (FC → device) |
| 3   | RX     | UART receive (device → FC)  |
| 4   | GND    | Ground                      |

# AF-H7E Lite Pin Definition

## • UART 6

JST-GH 4P · SERIAL6 · UART4

| Pin | Signal | Function                    |
|-----|--------|-----------------------------|
| 1   | VCC    | 5 V output                  |
| 2   | TX     | UART transmit (FC → device) |
| 3   | RX     | UART receive (device → FC)  |
| 4   | GND    | Ground                      |

## • I2C A

JST-GH 4P · I2C2

| Pin | Signal | Function                                                |
|-----|--------|---------------------------------------------------------|
| 1   | VCC    | 5 V output                                              |
| 2   | SCL    | I2C clock — external compass, rangefinder, airspeed ... |
| 3   | SDA    | I2C data                                                |
| 4   | GND    | Ground                                                  |

# AF-H7E Lite Pin Definition

## • I2C B

JST-GH 4P · I2C3

| Pin | Signal | Function                                                |
|-----|--------|---------------------------------------------------------|
| 1   | VCC    | 5 V output                                              |
| 2   | SCL    | I2C clock — external compass, rangefinder, airspeed ... |
| 3   | SDA    | I2C data                                                |
| 4   | GND    | Ground                                                  |

## • CAN 1

JST-GH 4P · CAN1 · DroneCAN

| Pin | Signal | Function                                          |
|-----|--------|---------------------------------------------------|
| 1   | VCC    | 5 V output to CAN peripherals (not a power input) |
| 2   | CAN_H  | CAN bus high                                      |
| 3   | CAN_L  | CAN bus low                                       |
| 4   | GND    | Ground                                            |

# AF-H7E Lite Pin Definition

## • CAN 2

JST-GH 4P · CAN2 · DroneCAN

| Pin | Signal | Function                                          |
|-----|--------|---------------------------------------------------|
| 1   | VCC    | 5 V output to CAN peripherals (not a power input) |
| 2   | CAN_H  | CAN bus high                                      |
| 3   | CAN_L  | CAN bus low                                       |
| 4   | GND    | Ground                                            |

## • RC IN

JST-GH 5P · rear edge · RC input · SBUS / PPM / DSM (auto-detected)

| Pin | Signal  | Function                                                                            |
|-----|---------|-------------------------------------------------------------------------------------|
| 1   | VCC     | 5 V receiver supply                                                                 |
| 2   | RC_IN   | SBUS / PPM / DSM receiver input (protocol auto-detected)                            |
| 3   | RSSI    | Analog RSSI input (0–3.3 V)                                                         |
| 4   | VCC_3V3 | Switched 3.3 V supply for DSM / Spektrum satellite receivers (power-cycled to bind) |
| 5   | GND     | Ground                                                                              |

# AF-H7E Lite Pin Definition

## • ETHERNET

JST-GH 4P · 100BASE-T

| Pin | Signal | Function                 |
|-----|--------|--------------------------|
| 1   | TX+    | Ethernet transmit pair + |
| 2   | TX-    | Ethernet transmit pair - |
| 3   | RX+    | Ethernet receive pair +  |
| 4   | RX-    | Ethernet receive pair -  |

## • DEBUG

JST-SH 6P · Pixhawk Debug Mini · SERIAL7 · USART3 console

| Pin | Signal     | Function                                                |
|-----|------------|---------------------------------------------------------|
| 1   | VREF       | 3.3 V reference for the debug probe                     |
| 2   | CONSOLE_TX | Debug console transmit (FC → probe)                     |
| 3   | CONSOLE_RX | Debug console receive (probe → FC)                      |
| 4   | SWDIO      | SWD data — bootloader programming and firmware recovery |
| 5   | SWCLK      | SWD clock                                               |
| 6   | GND        | Ground                                                  |

# AF-H7E Lite Pin Definition

## • USB

USB Type-C · SERIAL0 · USB 2.0 Full Speed

| Pin              | Signal    | Function                                                      |
|------------------|-----------|---------------------------------------------------------------|
| A4 B4<br>A9 B9   | VBUS      | 5 V from USB — configuration and firmware update on the bench |
| A6 B6            | D+        | USB data +                                                    |
| A7 B7            | D-        | USB data -                                                    |
| A5 B5            | CC1 / CC2 | Configuration channel (device role)                           |
| A1 B1<br>A12 B12 | GND       | Ground                                                        |

## • PWM OUT M1-M12

2.54 mm 3 × 13 pin header · columns 1-12 · Main outputs 1-12

| Pin | Signal     | Function                                                                                  |
|-----|------------|-------------------------------------------------------------------------------------------|
| S   | M1 ... M12 | Motor / servo output signal, one per channel (PWM · OneShot · DShot; M7-M8 without DShot) |
| +   | V_SERVO    | Servo rail, bussed across all channels — supplied externally (e.g. BEC)                   |
| -   | GND        | Ground                                                                                    |

# AF-H7E Lite Pin Definition

## • SB (SBUS OUT)

2.54 mm 3 × 13 pin header · column 13, next to M12 · SERIAL8  
· USART6 · SBus servo out

| Pin | Signal   | Function                                                                                                         |
|-----|----------|------------------------------------------------------------------------------------------------------------------|
| S   | SBUS_OUT | SBUS output — servo channels 1–16 on one wire (SERIAL8_PROTOCOL 15, SERIAL8_OPTIONS 2 inverts TX inside the MCU) |
| +   | V_SERVO  | Servo rail, bussed across all channels — supplied externally (e.g. BEC)                                          |
| –   | GND      | Ground                                                                                                           |

# AF-H7E Lite Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set `AHRS_ORIENTATION`.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 4.75 – 5.7 V (Rated 5 V).
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

# AF-H7E Lite Firmware

## • Available Firmware

| Type                                                         | Version | Date       |
|--------------------------------------------------------------|---------|------------|
| ArduPilot Copter (.apj package)                              | 0.1.0   | 2026-09-17 |
| ArduPilot Copter - Bootloader + App (merged HEX / DFU / SWD) | 0.1.0   | 2026-09-17 |
| ArduPilot Plane (.apj package)                               | 0.1.0   | 2026-09-17 |
| ArduPilot Plane - Bootloader + App (merged HEX / DFU / SWD)  | 0.1.0   | 2026-09-17 |

Download files from the product page (QR).

# AF-H7E Lite Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Product in development — the pin definition may change before release; check the latest catalog.
- Supply the PWM servo rail (+) from an external BEC.
- Firmware v0.1.0 is a preliminary, build-verified release. Copter and Plane share board ID 6207 — pick the image by file name.

# AF-H7E Lite Safety Notes



<Detailed Manual>

**novaX**